

Craig M. Weeks

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EDUCATION

Carnegie Mellon University

Doctor of Philosophy in Mechanical Engineering | GPA: 4.0/4.0

Master of Science in Mechanical Engineering

Pittsburgh, PA

May 2026

May 2025

Oregon State University

Bachelor of Science in Mechanical Engineering | GPA: 3.92/4.0

Minors in Aerospace Engineering and Computer Science

Corvallis, OR

September 2022

SKILLS

Software: MATLAB, Python, ANSYS, OpenFOAM, FLOW3D, SolidWorks, OnShape, Linux/HPC Computing

Hardware: Metal additive manufacturing, FDM 3D printing, Mechanical assembly, Machining

Analysis: Heat transfer, Computational fluid dynamics, Thermal imaging, Applied machine learning

PhD RESEARCH

Carnegie Mellon University

Spectral Observation and Simulation of Metal Vapor Plumes

Pittsburgh, PA

May 2025 - Present

- Analyzed multimodal spectral data on metallic vapor plumes to ascertain vapor interaction with high power laser irradiation and develop a physically based model of laser-based evaporation of metals
- Identified spectral signatures of ionizing titanium vapor atoms and vapor condensate regions with upwards of 20% light attenuation
- Created a multiphysics CFD model of vapor plume evaporation with a custom nucleation submodel and laser-vapor interaction for use in quantification of laser beam attenuation and additive manufacturing process planning development (**In Progress**)

Hybrid Ratiometric and Illuminated Process Monitoring

October 2024 - Present

- Co-developed a hybrid illuminated and ratiometric process monitoring technique to determine cooling rates in additive manufacturing processes by combining thermal gradients and morphological data obtained via laser reflections
- Developed a machine learning UNet model with >85% F1 Score to efficiently segment illuminated image video frames
- Reported temporally and spatially varying solidification cooling rates in 316L stainless steel melt pools between 170,000 and 3,000,000 K/s that demonstrate process phenomena and provide a foundation for process monitoring correlations

Simulated Imaging of L-PBF Melt Pools

July 2024 – March 2025

- Created an algorithm to simulate ratiometric thermal imaging on reflecting concave surfaces using radiation heat transfer physics and imported CFD data for understanding high temperature thermal distributions in liquid metals
- Reproduced temperature distributions to within 4-6% of experimental values and provided imaging recommendations for experimentalists to use when imaging

Multiphysics CFD Melt Pool Simulation

September 2022 – November 2024

- Developed and validated multiphysics CFD models of laser powder bed fusion (L-PBF) additive manufacturing in OpenFOAM® and FLOW3D® to study melt pool dynamics and evaporation
- Simulated L-PBF vapor plumes and found vapor velocities on the order of 100x greater than melt pool flow velocities

PUBLICATIONS

- Weeks, C. M.,** Myers, A. J., Malen, J. A., & Singh, S. (2025). Validation of OpenFOAM modeling of additive manufacturing melt pool dynamics against geometric and thermal experiments. *Journal of Manufacturing Processes*, 152, 237–249.
- Quirarte, G., Myers, A. J., Gourley, A., **Weeks, C. M.,** Reeja-Jayan, B., Beuth, J., & Malen, J. A. (2025). High speed thermal imaging and modeling of laser powder bed fusion manufactured WC–Ni cemented carbides. *Additive Manufacturing*, 110, 104913.
- Weeks, C. M.,** Malen, J. A., & Singh, S. (2025). Resolving Experimental Artifacts in Thermal Imaging of Laser Powder Bed Fusion Melt Pools. (in review)

CONFERENCES/PRESENTATIONS

- Weeks, Craig.** “Interpreting Temperature Distributions and Cooling Rates in Laser Powder Bed Fusion Through Surface Geometry and Illuminated Imaging.” Materials Science and Technology Technical Meeting and Exhibition, Columbus, OH, 1 October 2025, *Conference Presentation*
- Weeks, Craig.** “Simulation and Validation of Melt Pool Physics in the L-PBF Additive Manufacturing Process.” Materials Science and Technology Technical Meeting and Exhibition, Pittsburgh, PA, 8 October 2024, *Conference Presentation*
- Weeks, Craig.** “Numerical Simulation of Melt Pool Physics in Metal Additive Manufacturing Processes.” Solid Freeform Fabrication Symposium, Austin, TX, 13 August 2024, *Conference Presentation*
- Weeks, Craig.** “Numerical Simulation of Melt Pool Physics in Metal Additive Manufacturing Processes.” ASTFE Thermal and Fluids Engineering Conference, Corvallis, OR, 22 April 2024, *Conference Presentation*
- Weeks, Craig.** “Numerical Simulation of Melt Pool Physics in Metal Additive Manufacturing Processes.” Materials Science and Technology Technical Meeting and Exhibition, Columbus, OH, 2 October 2023, *Conference Presentation*
- Weeks, Craig.** “Hardware Design for an Efficient Inverter used in Electric Aircraft Power Systems.” Oregon Space Grant Consortium Student Symposium, Corvallis, OR, 13 November 2020, *Conference Presentation*

TEACHING ASSISTANT EXPERIENCE

- Carnegie Mellon University** Pittsburgh, PA
- Lead Teaching Assistant / 24-321 Thermal Fluids Experimentation January 2024 – May 2025
- Graded student homework and laboratory assignments
 - Facilitated online office hours and assisted students outside of classroom hours
 - Led undergraduate lab sessions and acted autonomously to respond to student questions and requests

AWARDS AND HONORS

- National Defense Science and Engineering Graduate Fellow** April 2023
- Oregon State University George and Evelyn Lundstrom Scholarship** September 2021
- National Merit Scholar** March 2018

VOLUNTEER EXPERIENCE

- Volunteer Instructor,** CMU Gelfand Outreach Center – Pittsburgh, PA July 2023 – July 2025
- Internship Peer Mentor,** NASA Glenn Research Center – Cleveland, OH June 2020 – August 2021

RELEVANT COURSEWORK

Graduate: 24-718 Computational Fluid Dynamics, 24-623 Molecular Simulation of Materials, 24-789 Intermediate Deep Learning for Engineers, 24-730 Advanced Heat Transfer

Undergraduate: ME312 Thermodynamics, ME332 Heat Transfer, ME461 Gas Dynamics, ME430 Systems Dynamics and Control, CS325 Analysis of Algorithms